

Field theory is very hard, Galois theory is the gymnastics of recovering information about fields from the much easier linear algebra, and finite groups acting on finite sets.

Definition 0.0.1. The **degree** of a field extension $K \subset L$ is $[L : K] := \dim_K L$ as a K -vector space. A field extension is called finite if its degree is.

From now on, unless otherwise said, all field extensions are finite.

Proposition 0.0.2 (Tower law). If $K \subset F \subset L$ is a tower of extensions, then $[L : K] = [L : F][F : K]$.

Proof. If $B = \{b_1, \dots, b_n\}$ and $C = \{c_1, \dots, c_m\}$ is a K -basis for F and an F -basis for L respectively, then $\{b_i c_j \mid 1 \leq i \leq n, 1 \leq j \leq m\}$ is a K -basis for L . $\square$

All field homomorphisms $f : K \rightarrow L$ are injective, as the kernel is a proper ideal of K , so instead of $\text{Hom}(K, L)$ I'll write $\text{Emb}(K, L)$. Now by the first isomorphism theorem, if $f : K \rightarrow K$ is a field *endomorphism*, it must be an isomorphism, so I'll write $\text{Aut}(K)$ for $\text{Emb}(K, K)$.

Definition 0.0.3. If $K \subset F, L$ are field extensions, then the set of K -embeddings from F to L is

$$\text{Emb}_K(F, L) = \{f \in \text{Emb}(F, L) \mid f|_K = \text{id}_K\}$$

In Galois theory, K is almost always the prime subfield, and so this distinction often doesn't matter. When there are multiple embeddings $K \rightarrow F, L$, by writing $K \subset F, L$ I am choosing a canonical embedding, and specifying some $f \in \text{Emb}_K(F, L)$ is asking for this map to commute with the canonical embeddings, in which case things like the cardinality of this set may depend on my choice of canonical embedding.

I will avoid this complication by almost always having a fixed *ground field* k of which every field K is an extension, so I am really working in the category of finite field extensions $k \subset K$. Here k is the special initial object, and I will try to state all propositions that don't require this, without using k as an object.

Proposition 0.0.4 (A2). If $K \subset L$ is a field extension, then there is a *right* action of $\text{Aut}_k(K)$ on $\text{Emb}_k(K, L)$ by composition, and this action is free.

Proof. As all $f \in \text{Emb}_k(K, L)$ are injective, if there exists $\sigma \in \text{Aut}_k(K)$ with $f\sigma = f$, then $\sigma = \text{id}_K$. $\square$

0.1 Categories

Proposition 0.1.1 (A4). For all finite extensions $K \subset F, L$ there exists a diagram:

$$\begin{array}{ccc} K & \subset & F \\ \cap & & \cap \\ L & \subset & \Omega \end{array}$$

Proof. Go by induction on $[F : K]$, the base case has $F = K$ where taking $\Omega = L$ works.

Now suppose $[F : K] > 1$, there will exist some $a \in F \setminus K$, which will have minimal polynomial $f \in K[X]$, let L' be the splitting field of $f \in L[X]$, then

$$\text{Emb}_k(k(a), L') = \{\text{roots of } f \text{ in } L'\} \neq \emptyset$$

and we can form the daigram:

$$\begin{array}{ccccc} K & \subset & K(a) & \subset & F \\ \cap & & \cap & & \\ L & \subset & L' & & \end{array}$$

where, as $K \neq K(a)$, the tower law says $[F : K(a)] < [F : K]$. $\square$

This looks like a pushout, but because we are asking for canonical embeddings $F, L \subset \Omega$ there is no universal property.

It's an exercise on one of the problem sheets to show fibered products, which are given by intersections, and "framed" pushouts, which are pushouts in the category of subfields of some Ω , and given by products, exist.

Definition 0.1.2. $k \subset K$ is a **normal extension** if for all $K \subset \Omega$ the action $\text{Emb}_k(K, \Omega) \curvearrowright \text{Aut}_k(K)$ is transitive, i.e. every $x : K \hookrightarrow \Omega$ is just the canonical inclusion of $\sigma(K)$ into Ω .

Lemma 0.1.3. If $k \subset K$ is normal, and $k \subset F \subset K$ is an intermediate field, then $F \subset K$ is normal.

Proof. Given $x, y \in \text{Emb}_F(K, \Omega) \subseteq \text{Emb}_k(K, \Omega)$ there exists some $\sigma \in \text{Aut}_k(K)$ such that $x = y\sigma$, this σ must be in $\text{Aut}_F(K)$. $\square$

Proposition 0.1.4. Let $k \subset K$ be normal and $k \subset F \subset K \subset \Omega$ be a tower, then the natural restriction:

$$\text{Emb}_k(K, \Omega) \rightarrow \text{Emb}_k(F, \Omega)$$

is surjective, i.e. any embedding $x : F \rightarrow \Omega$ extends to some $\tilde{x} : K \rightarrow \Omega$.

All components of a normal extension present in a bigger field still sit inside a copy of that normal extension.

Proof. For any $x \in \text{Emb}_k(F, \Omega)$, by (A4) I can form the following diagram:

$$\begin{array}{ccccc} & & \Omega & \xrightarrow{y} & \Omega' \\ & & \uparrow x & \nwarrow \tilde{x} & \uparrow z \\ k & \subset & F & \subset & K \end{array}$$

but as $k \subset K$ is normal, I must be able to write $z : K \rightarrow \Omega'$ as $y \circ \sigma$ for some $\sigma \in \text{Aut}_k(K)$. This σ (composed with the inclusion $K \subset \Omega$) is exactly the required $\tilde{x}$. $\square$

From this, it is immediately obvious that the natural injective composition with $K \subset \Omega$:

$$\text{Emb}_k(F, K) \rightarrow \text{Emb}_k(F, \Omega)$$

is also surjective.

Proposition 0.1.5 (A1). For all extensions $k \subset K, L$ the set $\text{Emb}_k(K, L)$ has cardinality at most $[K : k]$.

Proof. Once again going by induction on $[K : k]$ where the base case is trivial.

Supposing $[K : k] > 1$, pick $\alpha \in K \setminus k$ and consider the restriction map:

$$\rho : \text{Emb}_k(K, L) \rightarrow \text{Emb}_k(k(\alpha), L)$$

Now for any $x \in \text{Emb}_k(k(\alpha), L)$, the pre-image $\rho^{-1}(x) = \text{Emb}_x(K, L)$ has cardinality $\leq [K : k(\alpha)]$, so I can consider the sum:

$$|\text{Emb}_k(K, L)| = \sum_{x \in \text{Emb}_k(k(\alpha), L)} |\rho^{-1}(x)| \leq [k(\alpha) : k][K : k(\alpha)] = [K : k] \quad \square$$

Lemma 0.1.6. For an extension $k \subset K$ and $H \leq \text{Aut}(K/k)$ a finite subgroup, $[K : K^H] \leq |H|$.

Proof. If $G = \{\sigma_1, \dots, \sigma_n\}$, we need to show all sets $\{a_1, \dots, a_{n+1}\} \subset L$ are linearly dependent over K^H . Let $\{a_1, \dots, a_k\}$ consider the vectors:

$$\mathbf{a}_1 = \begin{pmatrix} \sigma_1(a_1) \\ \vdots \\ \sigma_n(a_1) \end{pmatrix}, \dots, \mathbf{a}_{n+1} = \begin{pmatrix} \sigma_1(a_{n+1}) \\ \vdots \\ \sigma_n(a_{n+1}) \end{pmatrix}$$

and wlog let $\{\mathbf{a}_1, \dots, \mathbf{a}_k\}$ be the smallest nontrivial linearly dependent set, so we have $x_i \neq 0$ such that $x_1 \mathbf{a}_1 + \dots + x_k \mathbf{a}_k = 0$, wlog take $x_1 = 1$. Applying some $\sigma \in \text{Aut}(K/k)$ will permute the rows of this linear system, but will always keep x_1 the same, this must always be the same system, otherwise we could subtract and find one of strictly smaller size. Thus all the $x_i \in K^H$ and equation corresponding to $\sigma_i = e$ is as required. $\square$

Corollary 0.1.7 (A5). Under the same assumptions, $[K : K^H] = |H|$, and the obvious inclusion $H \subset \text{Aut}(K/K^H)$ is an isomorphism.

Proof. (A1) already says $|\text{Aut}(K/K^H)| \leq [K : K^H]$, by the previous lemma we know $[K : K^H] \leq |H|$, and by assumption we know $|H| \leq |\text{Aut}(K/K^H)|$. $\square$

Definition 0.1.8. The extension $k \subset K$ is **separable** if for all towers of subfield:

$$k \subset K_1 \subsetneq K_2 \subset K$$

there exists some field Ω with at least two distinct K_1 -embeddings $x, y : K_2 \rightarrow \Omega$.

Lemma 0.1.9. If $k \subset K$ is separable, and $k \subset F \subset K$ is a tower, then $k \subset F$ and $F \subset K$ are separable.

Proof. Any pair of intermediate field of $k \subset F$ or $F \subset K$ are also intermediate field of $k \subset K$. $\square$

Theorem 0.1.10 (Fundamental theorem of Galois theory). If $k \subset K$ is Galois (normal and separable) then there is a bijective, order-reversing map between field and subgroups as follows:

$$H \leq G \mapsto K^H \quad k \subset F \subset K \mapsto \text{Aut}(K/F)$$

Proof. We know the images are fields and subgroups respectively, and the fact that these maps reverse the inclusion ordering is obvious. $H = \text{Aut}(K/K^H)$ is exactly (A5). As all $k \subset F \subset K$ are also Galois, we just have to prove $k' := K^{\text{Gal}(K/k)} = k$ we will go by showing that for all $k \subset \Omega$: $\text{Emb}_k(k', \Omega)$ is just the inclusion $k' \subset K \subset \Omega$, where separability now implies the final result. Consider the tower:

$$k \subset k' \subset K \subset \Omega$$

we know any $\text{Emb}_k(k', \Omega)$ is the restriction of some $\text{Emb}_k(K, \Omega)$ but as $k \subset K$ is normal, this corresponds to composing the inclusion $K \subset \Omega$ with some $\sigma \in \text{Gal}(K/k)$, by construction this is just $k' \subset K \subset \Omega$. $\square$

We now want to translate the categorical definitions of normal and separable extensions into field theoretic statements.

0.2 Normal extensions

Theorem 0.2.1. For a finite extension $k \subset K$ tfae:

1. For all irreducible $f \in k[X]$, either f has no roots in K or f splits completely in K ;¹
2. There exists some $f \in k[X]$ such that K is the splitting field of f over k ;
3. $k \subset K$ is normal.

Proof. (1 $\Rightarrow$ 2) If $K = k(a_1, \dots, a_n)$ and f_i is the minimal polynomial of a_i over k respectively, K is clearly the splitting field of $f_1 \cdots f_n$,

(2 $\Rightarrow$ 3) If $k \subset K$ is the splitting field of f with root $a \in K$, then for any $K \subset \Omega$ and $x : K \rightarrow \Omega$, $x(a)$ will always be a root of f in Ω and as K is generated by all the roots of f , $x(K) \subset K \subset \Omega$.

(3 $\Rightarrow$ 1) Now if $k \subset K$ is normal, for any irreducible $f \in k[X]$ let $K \subset \Omega$ be the splitting field of f over K . If f has any root $a \in K$ then we can form the tower $k \subset k(a) \subset K \subset \Omega$ giving the bijection:

$$\{\text{roots of } f \text{ in } \Omega\} = \text{Emb}_k(k(a), \Omega) = \text{Emb}_k(k(a), K) = \{\text{roots of } f \text{ in } K\} \quad \square$$

Lemma 0.2.2. For all $k \subset K$ there exists some $K \subset L$ such that $k \subset L$ is normal.

Proof. Let $K = k(a_1, \dots, a_n)$, with f_i the minimal polynomial of each a_i over k respectively, now let L be the splitting field of $f_1 \cdots f_n$ over K , and thus also over k . $\square$

¹This is a standard definition of a normal extension

0.3 Separable extensions

Definition 0.3.1. For $k \subset K$ define the **separable degree**

$$[K : k]_s := |\text{Emb}_k(K, \Omega)|$$

where $k \subset K \subset \Omega$ is normal extension that exists by the previous lemma.

Lemma 0.3.2. The separable degree $[K : k]_s$ does not depend on the choice of Ω .

Proof. Let Ω_1, Ω_2 be such normal extensions of k , then by (A4) let Ω' be a field containing both of them, we know any embedding $K \rightarrow \Omega'$ must send K to the normal subfield containing it, i.e.

$$\text{Emb}_k(K, \Omega_1) = \text{Emb}_k(K, \Omega) = \text{Emb}_k(K, \Omega_2) \quad \square$$

Separable extensions are now exactly those $k \subset K$ such that for all towers $k \subset K_1 \subset K_2 \subset K$, if $[K_1 : K_2]_s = 1$, then $K_1 = K_2$.

Proposition 0.3.3. For a tower $k \subset K \subset L$ we have

$$[L : k]_s = [L : K]_s [K : k]_s$$

Proof. Take $L \subset \Omega$ such that $k \subset \Omega$ is normal, we know the restriction:

$$\text{Emb}_k(\Omega, \Omega) \rightarrow \text{Emb}_k(L, \Omega) \xrightarrow{\rho} \text{Emb}_k(K, \Omega)$$

is surjective, so we can count

$$[L : k]_s := |\text{Emb}_k(L, \Omega)| = \sum_{x \in \text{Emb}_k(K, \Omega)} |\rho^{-1}(x)| = [K : k]_s [L : K]_s$$

as the previous lemma says the size of $\rho^{-1}(x) = \text{Emb}_K(L, \Omega)$ doesn't depend on the choice $x : K \rightarrow \Omega$. $\square$

Definition 0.3.4. A polynomial $f \in K[X]$ is **separable** if it has $n = \deg(f)$ distinct roots in its splitting field. An element $a \in K$ is then called separable if its minimal polynomial is.

Proposition 0.3.5. An irreducible polynomial $f \in K[X]$ is inseparable iff K has characteristic $p > 0$ and there exists some irreducible $h \in K[X]$ with $f(X) = h(X^p)$.

Lemma 0.3.6. If $K \subset K(a)$ is separable, then a is separable over K .

Proof. Let $f \in K[X]$ be the minimal polynomial of a over K , and suppose it isn't separable, then we have $f(X) = h(X^p)$ for some irreducible h , and $b = a^p$ is a root of h , so we can form the tower $K \subset K_1 = K(b) \subset K_2 = K(a)$ where, by the tower law

$$p \deg(h) = \deg(f) = [K_2 : K] = [K_2 : K_1][K_1 : K] = [K_2 : K_1] \deg(h)$$

so $[K_2 : K_1] = p$ and $(X - a)^p = X^p - b \in K_1[X]$ is the minimal polynomial of a over K_1 , so $[K_2 : K_1]_s = 1$, and thus $K \subset K(a)$ is inseparable. $\square$

Theorem 0.3.7. A finite extension $k \subset K$ is separable iff $[K : k]_s = [K : k]$.

Proof. ($\Leftarrow$) If $[K : k]_s = [K : k]$, then by applying two tower laws to

$$k \subset K_1 \subset K_2 \subset K$$

and by (A1), $[- : -]_s \leq [- : -]$, if $[K_2 : K_1]_s = 1$, so if $[K_2 : K_1]_s = 1$, we get

$$[K : K_2][K_2 : K_1][K_1 : k] = [K : K_2]_s [K_1 : k]_s \leq [K : K_2][K_1 : k]$$

so $[K_2 : K_1] = 1$ and $K_1 = K_2$.

($\Rightarrow$) Now, if $k \subset K$ is separable, pick $a \in K \setminus k$ and consider the tower $k \subset k(a) \subset K$, we have $[k : k(a)]$ separable, so by the previous lemma we know a is separable over k , so

$$[k(a) : k] = \deg(f) = |\{\text{roots of } f\}| = [k(a) : k]_s$$

and as $[K : k(a)] < [K : k]$, we may inductively assume $[K : k(a)]_s = [K : k(a)]$, and the tower law finished the proof. $\square$

Corollary 0.3.8. The following list of statements immediately hold:

1. For all towers $k \subset K \subset L$, if $k \subset K$ and $K \subset L$ are separable, then $k \subset L$ is separable;
2. $k \subset K$ is separable iff every $a \in K$ is separable over k ;
3. if a is separable over k , then $k \subset k(a)$ is separable;
4. if $k \subset K$ is the splitting field of a separable polynomial, then $k \subset K$ is separable.

0.4 Primitive element theorem

This is Stacks project lemma 9.19.1, and very much not categorical.

Proposition 0.4.1. For all finite field extensions $K \subset L$, tfae:

1. L is **simple** over K , i.e. there exists some $\alpha \in L$ such that $L = K(\alpha)$;
2. there are finitely many intermediary subfields $K \subset F \subset L$.

Proof. ($\Rightarrow$) suppose we have a subfield $K \subset F \subset K(\alpha)$ with f the minimal polynomial of α over F , then $\deg(f) = [K(\alpha) : F]$ and if $f = x^d + a_{d-1}x^{d-1} + \cdots + a_0$, then $K(a_0, \dots, a_{d-1}) \subset F$ and they have the same degree in $K(\alpha)$, so are equal. All such f must be one of the finitely many factors of the minimal polynomial of α in K when viewed in $K(\alpha)$.

($\Leftarrow$) If we are working with finite fields then we can always find primitive elements as generators of the cyclic groups $L^\times$. Now assume K is infinite, and has only finitely many subfields $K_1, \dots, K_n$, these are all proper subspaces so we can find $\alpha \notin K_i$ for all i , this is our primitive element. $\square$

Proposition 0.4.2. If $K \subset L$ is separable, then the above conditions hold.

Proof. Let $k \subset L \subset \Omega$ be a normal extension—so $K \subset L \subset \Omega$ is as well—and consider an enumeration of the finite set $\text{Emb}_K(L, \Omega) = \{\sigma_1, \dots, \sigma_n\}$, if $K \subset L$ is separable, then $[L : K] = n$ and for all $i \neq j$ we have $V_{ij} = \ker(\sigma_i - \sigma_j) \subsetneq L$ from which we can find a primitive element as in the previous proposition. $\square$

This proof doesn't provide a reasonable way to compute the primitive element though.